

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

V_{emf} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}

Motional EMF *loop moves, B static*

V_{emf} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}

General (flux form) *always works*

V_{emf} = - \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}

N-turn coil *transformer, N turns*

V_{emf} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \quad \lambda = N\Phi

Uniform B shortcut *B same everywhere on loop*

V_{emf} = - \frac{\partial B}{\partial t} \cdot A

Long wire + rect loop *wire near loop, r: a \to b, height c*

V_{emf} = - \int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz

MAXWELL'S EQUATIONS

Name	Differential form
Faraday	$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$
Gauss (E)	$\nabla \cdot \mathbf{D} = \rho_v$
Gauss (B)	$\nabla \cdot \mathbf{B} = 0$

D = εE; B = μH; source-free: ρ_v = 0, J = 0

SURFACE ELEMENT ds

ds = n da (RHR gives n: curl fingers in loop direction, thumb = n)

Loop plane	ds
x-y	z dx dy or z r dr dφ
y-z	x dy dz
x-z	y dx dz
coplanar / toroid	φ dr dz

Uniform B: ∫∫ da = A (just the loop area)

DOT PRODUCTS â · b̂

Same direction = 1; perpendicular = 0

	â	ŷ	ẑ	ŕ	φ̂
â	1	0	0	0	0
ŷ	0	1	0	0	0
ẑ	0	0	1	0	0
φ̂	0	0	0	0	1

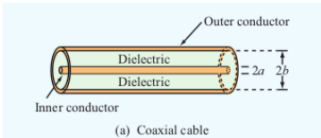
CROSS PRODUCT A × B (determinant)

A × B = |â Ax Ay Az; Bx By Bz| = (AyBz - AzBy)â - (AxBz - AzBx)ŷ + (AxB y - AyBx)ẑ
Also used for curl:

∇ × E = |∂x ∂y ∂z; Ex Ey Ez|

Plane wave (∂x = ∂y = 0): only ∂z row survives.

COAXIAL CABLE — AC RESISTANCE



R' = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) (\Omega/m)

a: inner radius; b: outer radius; R_s = \sqrt{\pi f \mu \sigma}

FIELDS

E — electric field (V/m)
D = εE — elec. flux density (C/m²)
B — mag. flux density (T)
H — mag. field intensity (A/m)
ρ_v — charge density (C/m³)
ρ_v0 — initial charge density
Φ = ∫_S B · ds — flux (Wb)
λ = NΦ — flux linkage (Wb)

CURRENTS

J_c = σE — cond. density (A/m²)
J_d = ε ∂E / ∂t — disp. density
I_c = J_c · A — cond. current (A)
I_d = J_d · A — disp. current (A)
R = d / (σA) — resistance (Ω)
C = ε_r ε_0 A / d — capacitance (F)
τ_r = ε_r ε_0 / σ — relax. time (s)

MATERIALS

μ_0 = 4π × 10⁻⁷ H/m
μ_r — relative permeability
μ = μ_r μ_0 — permeability
ε_0 = 8.85 × 10⁻¹² F/m
ε_r — relative permittivity
ε = ε_r ε_0 — permittivity
σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
R — resistance (Ω)
C — capacitance (F)
ω = 2πf — angular freq (rad/s)
k = ω√μ ε — wave number (m⁻¹)
λ = 2π/k — wavelength (m)
c = 3 × 10⁸ m/s — speed of light

PHASOR & GEOMETRY

η = √μ/ε — intrinsic imp. (Ω)
η_0 = 120π ≈ 377 Ω (free space)
k/(ωμ) = 1/η
j — imaginary unit (j² = -1)
E — phasor (drop e^{jωt})
A — area (m²); d — sep. (m)
N — turns; r — radial dist. (m)

B FIELDS

Infinite wire *single wire carrying I, N = 1*

B = φ̂ \frac{\mu_0 I}{2\pi r}

Inside toroid *coil of N turns carrying I*

B = φ̂ \frac{\mu N I}{2\pi r} \quad \mu = \mu_r \mu_0

N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire ⇒ N = 1, drop it. N in V_{emf} = -NdΦ/dt is the receiver coil turns.

∂B/∂t TABLE

B(t)	∂B/∂t
B_0 sin(ωt)	B_0 ω cos(ωt)
B_0 cos(ωt)	-B_0 ω sin(ωt)
B_0 e^{-αt}	-α B_0 e^{-αt}
B_0 (const)	0

SIGN → CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	- → + terminal	-φ̂ (CW)
> 0	+ → - terminal	+φ̂ (CCW)
= 0	—	none

Lenz: induced I always opposes the change in Φ.

ln INTEGRAL

B ∝ 1/r, loop spans r = a to r = b

\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a

\ln \frac{b}{a} = \ln b - \ln a

\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182

ROTATING LOOP — GENERATOR

$\Phi = AB_0 \cos(\omega t + \phi_0)$ $V_{emf} = AB_0 \omega \sin(\omega t + \phi_0)$ $I_{amp} = \frac{AB_0 \omega}{R} \quad \omega = 2\pi f = 2\pi \frac{rpm}{60}$ Common loop areas	
Shape	Area A
Square (side a)	a²
Rectangle (l × w)	lw
Circle (radius r)	πr²
Equil. triangle (side a)	\frac{\sqrt{3}}{4} a²
Any triangle	\frac{1}{2} bh

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10⁻¹²	kilo	k	10³
nano	n	10⁻⁹	mega	M	10⁶
micro	μ	10⁻⁶	giga	G	10⁹
milli	m	10⁻³	tera	T	10¹²

j IDENTITIES

Expression	Result	Expression	Result
j²	-1	j × (-j)	1
1/j	-j	-1/j	j
j³	-j	j⁴	1
e^{-jπ/2}	-j	e^{jπ/2}	j
e^{jπ}	-1	e^{j2π}	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, σ ≠ 0*

E = \frac{V(t)}{d} \quad J_c = \sigma E \quad I_c = J_c \cdot A = \frac{\sigma A}{d} V(t)

I_c = \frac{\sigma A}{d} V(t) \iff I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}

Current proportional to V directly → behaves like a resistor.

Displacement *capacitive, changing E*

J_d = \epsilon \frac{\partial E}{\partial t} \quad I_d = J_d \cdot A = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt}

I_d = \underbrace{\frac{\epsilon_r \epsilon_0 A}{d}}_C \frac{dV}{dt} \iff I = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}

Current proportional to dV/dt → behaves like a capacitor.

Lossy cap circuit *both σ and ε_r present*

R || C across V(t): I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}

CHARGE-CURRENT CONTINUITY

\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}

Conductor (steady): \nabla \cdot \mathbf{J} = 0 \Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{s} = 0 (KCL)

CHARGE RELAXATION

Decay *charge injected at t = 0*

\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma}

Find σ from decay data

\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD (E → H)

1. Key relation E(z, t) = Re{Ẽ(z) e^{jωt}}
2. Euler split e^{j(ωt-kz)} = e^{jωt} · e^{-jkz} strip e^{jωt}, keep e^{-jkz}

3. Time → phasor conversions
cos(ωt - kz) = Re{e^{j(ωt-kz)}} = Re{e^{jωt} · e^{-jkz}} → e^{-jkz}
sin(ωt - kz) = cos(ωt - kz - π/2) = Re{e^{j(ωt-kz-π/2)}} = Re{e^{-jπ/2} · e^{jωt} · e^{-jkz}} → -j e^{-jkz}

4. Faraday in phasor domain (plane wave: only ∂/∂z survives)

H̃ = - \frac{1}{j\omega\mu} \nabla \times Ẽ

5. Phasor → time domain H(z, t) = Re{H̃ e^{jωt}}
Re{e^{j(ωt-kz)}} = cos(ωt - kz)
Re{-j e^{j(ωt-kz)}} = sin(ωt - kz)

UNIT CIRCLE

Angle	Deg	cos	sin	Q
0	0	1	0	—
π/6	30	√3/2	1/2	1
π/4	45	√2/2	√2/2	1
π/3	60	1/2	√3/2	1
π/2	90	0	1	—
2π/3	120	-1/2	√3/2	2
3π/4	135	-√2/2	√2/2	2
5π/6	150	-√3/2	1/2	2
π	180	-1	0	—
7π/6	210	-√3/2	-1/2	3
5π/4	225	-√2/2	-√2/2	3
4π/3	240	-1/2	-√3/2	3
3π/2	270	0	-1	—
5π/3	300	1/2	-√3/2	4
7π/4	315	√2/2	-√2/2	4
11π/6	330	√3/2	-1/2	4
2π	360	1	0	—

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz
 $\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$

Phase velocity m/s ; nonmagnetic: $\mu_r = 1$
 $u_p = \frac{c}{\sqrt{\epsilon_r}} \quad c = 3 \times 10^8 \text{ m/s}$

Wavenumber k rad/m
 $k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$

Wavelength & impedance $m \text{ \& } \Omega$
 $\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \Omega$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E
 $\mathbf{E} = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0)\hat{e}$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space;
 $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t , position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. $\mathbf{E}$ along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.
 $\mathbf{H}$ from $\mathbf{E}$, $\hat{E}$, $\hat{H}$ form right-hand set
 $\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$

Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0
 $E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$
 $\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

LOSSLESS vs. LOSSY — E & H

	Lossless ($\sigma = 0$)	Lossy ($\sigma \neq 0$)
Impedance Prop.	$\eta = \sqrt{\mu/\epsilon}$ e^{-jkz}	$\eta_c = \eta_c e^{j\theta_\eta}$ $e^{-\alpha z}e^{-j\beta z}$
$\mathbf{E}(t)$	$\text{Re}\{\tilde{\mathbf{E}}e^{j\omega t}\}$	(same)
$\mathbf{H}(t)$	$\text{Re}\{\tilde{\mathbf{H}}e^{j\omega t}\}$	(same)
$\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$	$\frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$	$\frac{1}{\eta_c}(\hat{k} \times \tilde{\mathbf{E}})$
$\tilde{\mathbf{E}}$ from $\tilde{\mathbf{H}}$	$-\eta(\hat{k} \times \tilde{\mathbf{H}})$	$-\eta_c(\hat{k} \times \tilde{\mathbf{H}})$
Phase	in phase	$\mathbf{H}$ lags $\mathbf{E}$ by θ_η

Lossless shortcuts (wave in $+z$, $\mathbf{E}$ along $\hat{e}$):
 $\mathbf{E}(z, t) = E_0 \cos(\omega t - kz + \phi)\hat{e}$
 $\mathbf{H}(z, t) = \frac{E_0}{\eta} \cos(\omega t - kz + \phi)(\hat{k} \times \hat{e})$
 $\mathbf{H}(t)$ direct $\frac{1}{\eta}(\hat{k} \times \mathbf{E}(t))$
 $\mathbf{E}(t)$ direct $-\eta(\hat{k} \times \mathbf{H}(t))$

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k -axis $\Rightarrow +$ direction. Plus $\Rightarrow -$ direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k} \times \hat{k} = \mathbf{0}$; any $\hat{a} \times \hat{a} = \mathbf{0}$.
Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS ($\theta \equiv \omega t - kz$)

Input	Result	Input	Result
$\sin(\theta + \pi/2)$	$\cos \theta$	$\cos(\theta + \pi/2)$	$-\sin \theta$
$\sin(\theta - \pi/2)$	$-\cos \theta$	$\cos(\theta - \pi/2)$	$\sin \theta$
$\sin(\theta + \pi)$	$-\sin \theta$	$\cos(\theta + \pi)$	$-\cos \theta$
$\sin(\theta - \pi)$	$-\sin \theta$	$\cos(\theta - \pi)$	$-\cos \theta$

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
 $k = \omega\sqrt{\epsilon_r}/c$ — wavenumber (rad/m)
 $\lambda = 2\pi/k$ — wavelength (m)
 $u_p = c/\sqrt{\epsilon_r}$ — phase vel. (m/s)
 $\eta = \eta_0/\sqrt{\epsilon_r}$ — impedance (Ω)
 $c = 3 \times 10^8$ m/s
 $\eta_0 = 377 \Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
 $\epsilon'' = \sigma/\omega = 0 \Rightarrow \epsilon = \epsilon'$ (real)
 $\alpha = 0$; no attenuation
 $\eta = \eta_0/\sqrt{\epsilon_r}$ real (Ω)
 $\eta_0 = 377 \Omega$ (free space)
 $\mathbf{E}$ and $\mathbf{H}$ in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
 β — phase const (rad/m)
 $\gamma = \alpha + j\beta$ — prop. const (m⁻¹)
 η_c — complex impedance (Ω)
 $\epsilon'' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$ (F/m)
 μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
 $\mu = \mu_r \mu_0$; $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $\delta_s = 1/\alpha$ — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
 a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
 $-j = e^{-j\pi/2}$: $\cos \rightarrow \sin$
 $+j = e^{+j\pi/2}$: $\cos \rightarrow -\sin$

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
 $\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
 $\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
 $\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
 $\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
 $\arctan(24/7) \approx 73.7^\circ = 1.287$ rad

POLARIZATION — GENERAL WAVE

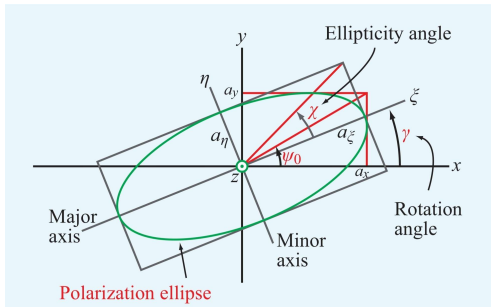


Figure 7-11 Polarization ellipse in the x - y plane with the wave traveling in the z direction (out of the page).

$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$
 a_x, a_y : bounding box half-widths (V/m); δ : phase diff (rad)
 $\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ (linear)
 $\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ (linear)
Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$
Rotation angle γ (major axis from $\hat{x}$) $\gamma \in [0, \pi)$
 $\tan(2\gamma) = \tan(2\psi_0) \cos \delta$
If $\text{sign}(\gamma) \neq \text{sign}(\delta)$: $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.
Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$
 $\sin(2\chi) = \sin(2\psi_0) \sin \delta$
Axial ratio $R = 1/|\tan \chi|$ major axis = $R \times$ minor; $R \geq 1$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0)$			
N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$\gamma = (2\gamma)/2$. $D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
Convert: $\theta_{\text{rad}} = \theta_0 \times \frac{\pi}{180}$; $\theta_0 = \theta_{\text{rad}} \times \frac{180}{\pi}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	$\text{Re}\{\cdot\} e^{j\omega t} e^{-\alpha z} e^{-j\beta z}$
$A (= Ae^{j\phi})$	$A \cos(\omega t - \beta z)$
$-jA (= Ae^{-j\pi/2})$	$A \sin(\omega t - \beta z)$
$+jA (= Ae^{+j\pi/2})$	$-A \sin(\omega t - \beta z)$
$-A (= Ae^{\pm j\pi})$	$-A \cos(\omega t - \beta z)$
$Ae^{j\phi}$ (general)	$A \cos(\omega t - \beta z + \phi)$
$A + jB (= Z e^{j\theta})$	$ Z \cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check z quadrant)

CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$
RHC $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} + \sin(\omega t - kz)\hat{y}]$
LHC $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} - \sin(\omega t - kz)\hat{y}]$

LOSSY MEDIA — CLASSIFICATION

Loss tangent

ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\epsilon' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\epsilon_r = \mu_r = 1$.

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\epsilon''/\epsilon')^2}$)

$\alpha = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X-1}$ Np/m, $\beta = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X+1}$ rad/m
 $\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}$ (m), $u_p = \frac{\omega}{\beta}$ (m/s)
 $\eta_c = \frac{\eta}{X^{1/2}} e^{j\theta_\eta}$ (Ω), $\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'} \text{ (rad)}$
Rect: $\eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta$, $|\eta_c| = \eta/X^{1/2}$
 $\eta = \eta_0/\sqrt{\epsilon_r}$; $\epsilon' = \epsilon_r \epsilon_0$.

SKIN DEPTH δ_s (m) — ANY LOSSY MEDIUM

General: $\delta_s = 1/\alpha$ $|\mathbf{E}(z)| = E_0 e^{-z/\delta_s}$; lossless: $\alpha = 0 \Rightarrow$ no decay
Low-loss ($\epsilon''/\epsilon' \ll 1$) $\delta_s \approx 2/(\sigma\eta)$
Good conductor ($\epsilon''/\epsilon' \gg 1$) $\delta_s \approx 1/\sqrt{\pi f \mu \sigma}$

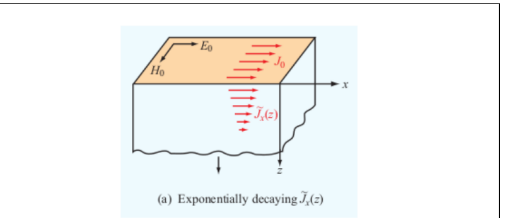
LOW-LOSS MEDIUM ($\epsilon''/\epsilon' \ll 1$)

$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma\eta}{2}$ (Np/m)
 $\beta \approx \omega \sqrt{\mu\epsilon'} = \frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), $u_p \approx \frac{c}{\sqrt{\epsilon_r}}$ (m/s)
 $\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta$ (Ω , $\angle \approx 0$)

GOOD CONDUCTOR ($\epsilon''/\epsilon' \gg 1$)

$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$ (Np/m, rad/m)
 $u_p \approx \sqrt{4\pi f / \mu \sigma}$ (m/s), $\lambda \approx u_p/f$ (m)
 $\eta_c \approx (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma}$ (Ω)

SURFACE RESISTANCE (GOOD CONDUCTOR)



$J_0 = \sigma E_0$ — current density at surface ($z=0$)
 $\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{x}$ — decays with depth
 $R_s = \sqrt{\pi f \mu / \sigma} = 1/(\sigma \delta_s)$ (Ω)
 $Z_s = (1+j)R_s$ — surface impedance
 $R_{AC} = R_s \ell/w$ — AC resistance of slab

AVERAGE POWER DENSITY (POYNTING)

General W/m^2
 $S_{av} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$
Lossless shortcut $|\mathbf{E}_0|^2 = a_x^2 + a_y^2 + a_z^2$
 $S_{av} = \frac{|\mathbf{E}_0|^2}{2\eta} \hat{k}$
Lossy medium $\eta_c = |\eta_c| e^{j\theta_\eta}$; wave in $+\hat{z}$
 $S_{av} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \hat{z}$
Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then S_{av} is a negative x -component.

DECIBELS & ATTENUATION

$G = 10 \log(P_1/P_2) = 20 \log(|E_1|/|E_2|)$ (dB)
 $A = -8.69 \alpha z$ (dB); α (dB/m) $\equiv 8.69 \alpha$ (Np/m)